

INTERACTION OF *LISTERIA MONOCYTOGENES* AND *STAPHYLOCOCCUS EPIDERMIDIS* IN DUAL SPECIES BIOFILMS

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Accepted for Publication March 28, 2010

ABSTRACT

The development of mixed biofilms of Listeria monocytogenes and Staphylococcus epidermidis was studied on two model surfaces. Growth tests at 30C for mono and mixed cultures of the two bacteria in defined Hsiang-Ning Tsai Medium, in tryptic soy broth (TSB) +1% glucose, TSB + 0.6% yeast extract (TSBYE) and TSBYE + 1% glucose was performed. The optimal growth was seen in TSBYE + 1% glucose, and hence, biofilm formation in this medium was studied using a polystyrene microtiter plate assay. Mono and mixed cultures with different ratios of the two bacteria were assayed for (1) biofilm strength, expressed as the amount of cell-bound crystal violet and (2) amount of matrix polysaccharide measured by ruthenium red staining. The mixture with a 40:60 ratio of L. monocytogenes to S. epidermidis yielded the highest values in both assays. The biofilm strength measured for this ratio indicated a synergistic interaction between the two bacterial species. Scanning electron microscopy of biofilms formed on cellulose membranes also revealed such an interaction.

PRACTICAL APPLICATIONS

Within food processing environments, bacterial biofilms are of concern, in particular if they contain foodborne pathogens such as *Listeria monocytogenes* which may (re-)contaminate food products. Our study shows that the

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intrinsically weak biofilm former *L. monocytogenes* strain EGD-e, can form more robust biofilms in association with bacteria like *Staphylococcus epidermidis*, which as such are harmless in food, but form strong biofilms. This suggests that also such synergistically acting bacteria should be monitored and controlled.

INTRODUCTION

Listeria monocytogenes and *Staphylococcus epidermidis* are two opportunistic human pathogens, causing major problems in the food sector and in the medical field. *L. monocytogenes* is a gram-positive, facultative anaerobic bacterium widespread in nature. It is remarkably robust, growing at temperatures between 2 and 45°C, at pH 5–9 and salt concentrations up to 12% (Juntilla *et al.* 1988; Van der Veen *et al.* 2008). It is transmitted by contaminated food (Schlech *et al.* 1983; Swaminathan and Gerner-Smidt 2007) and can cause a multifaceted, often fatal disease called listeriosis, particularly in children, pregnant women, elderly and immunocompromised persons (Wing and Gregory 2002). It has also been implicated in a number of gastroenteritis cases (Ooi and Lorber 2005). On the other hand, the coagulase-negative *S. epidermidis*, once regarded as harmless skin commensal or contaminant of clinical material, has emerged as a major nosocomial pathogen (Vuong and Otto 2002), being harmless only as long as it thrives on abiotic surfaces or on intact epithelia.

Almost all bacterial species, under appropriate conditions, are able to adhere to various abiotic and biotic surfaces (Dunne 2002). Given a sufficient supply of nutrients, the bacteria then eventually will form a biofilm on the respective surface (Costerton *et al.* 1995; Hall-Stodley *et al.* 2004; Jefferson 2004) by a complex process that is not completely understood. The initial adhesion of *L. monocytogenes* to inert surfaces seems to involve a homolog of the *Escherichia coli* Era protein (Auvray *et al.* 2007) and the biofilm-associated protein BapL (Jordan *et al.* 2008) as well as extracellular DNA (Harmsen *et al.* 2010). The role of flagella and motility in listerial adhesion and biofilm formation is controversial (Di Bonaventura *et al.* 2007; Lemon *et al.* 2007; Todhanakasem and Young 2008). For *S. epidermidis*, it has been described that the attachment to unmodified polymer surfaces is mediated by capsular polysaccharide (Tojo *et al.* 1988) and also by proteins like autolysin AtlE (Heilmann *et al.* 1997) and SSP (Veenstra *et al.* 1996). During the subsequent phase, bacteria become enmeshed in a mass of extracellular material that is firmly attached to the underlying surface. The composition of this matrix will be discussed later.

Biofilms can form on a wide range of inert surfaces like metal, rubber, glass and polymers as found on equipment and contact surfaces in food

processing facilities (Donlan 2002; Di Bonaventura *et al.* 2007). Biofilms of pathogenic bacteria, e.g., *L. monocytogenes*, in food manufacturing and processing facilities are of particular concern because they are difficult to eliminate (Fenlon *et al.* 1996; Norwood and Gilmour 2000; Robbins *et al.* 2005) and bacteria can easily be transferred to food products. Biofilms found in natural and food processing environments in most cases contain more than one bacterial species (Watnick and Kolter 2000; Hall-Stodley *et al.* 2004). This can also be anticipated for *L. monocytogenes* and *S. epidermidis* as both are found in food processing environments (Giovannaci *et al.* 1999; Di Bonaventura *et al.* 2007; Schlegelova *et al.* 2008). The former bacterium is contaminating via raw material or animal fecal matter, the latter one is a normal constituent of the skin microflora. Basically, *L. monocytogenes* in general and strain EGD-e in particular (Tresse *et al.* 2006) is a poor biofilm former when compared to *S. epidermidis* RP62A, the latter being a slime producing strong biofilm former (Sadovskaya *et al.* 2005). However, it can be assumed that when present together, a biofilm formed by the normally harmless *S. epidermidis* RP62A can serve in a synergistic way as a support for the pathogenic *L. monocytogenes* EGD-e. The latter strain was chosen because we assumed that any synergistic interaction would be more pronounced with a weak biofilm producer like *L. monocytogenes* EGD-e. Furthermore, it is a widely used reference strain for which a large number of mutants are available and it belongs to serovar 1/2a which is frequently found among *L. monocytogenes* in food processing plants (Giovannaci *et al.* 1999). *S. epidermidis* RP62A is a widely used biofilm-positive reference strain (Sadovskaya *et al.* 2005). For both strains, the complete genome sequences have been determined (Glaser *et al.* 2001; Gill *et al.* 2005), which will facilitate future molecular biological investigations. In this study, we attempted to investigate the possible interactions during biofilm formation in dual species cultures of *L. monocytogenes* EGD-e and *S. epidermidis* RP62A. As illustrated above, these two strains constitute a suitable model system for such studies.

MATERIALS AND METHODS

Organisms and Growth Media

The two bacterial isolates, namely *L. monocytogenes* EGD-e (serovar 1/2a) (ATCC BAA-679) and *S. epidermidis* (RP62A), were used for the study. *L. monocytogenes* was kindly provided by T. Chakraborty, University of Giessen, Germany and the latter by K. Ohlsen, University of Würzburg, Germany. All the media and chemicals used were purchased from Himedia Pvt. India Ltd., Mumbai. Single colonies from the stock cultures were

inoculated into tryptic soy broth (TSB) supplemented with 1% (w/v) glucose and 0.6% yeast extract (w/v) and incubated for 24 h at 37°C in an orbital shaker incubator (Thermo Scientific, Hamburg, Germany) at 190 rpm. These precultures were centrifuged ($5,000 \times g$ at 4°C for 10 min) and the pellets were washed twice with phosphate-buffered saline (PBS) pH 7.0 and resuspended in 5 mL of PBS.

Standardization of Growth Medium and Culture Conditions

Initial experiments were performed with different culture media, namely defined Hsiang-Ning Tsai Medium (HTM) supplemented with amino acids (Tsai and Hodgson 2003), TSB with 1% (w/v) glucose, TSB with 0.6% (w/v) yeast extract (TSBYE) and TSBYE with 1% glucose. Previously, we could show that biofilm formation by *L. monocytogenes* EGD-e was best at 30°C (Zameer *et al.* 2009); therefore, this temperature was used here. Bacterial growth was measured as optical density (OD) at 600 nm, using an Ultrospec 3100 spectrophotometer (Amersham Biosciences, Piscataway, NJ).

Cultivation of Biofilms

A resuspended preculture of either the individual species or mixed at a ratio of 20:80, 40:60, 50:50, 60:40 and 80:20 of *L. monocytogenes* and *S. epidermidis*, respectively, were further diluted to 5,000-fold (to approximately 5×10^6 cfu/mL) with TSBYE + 1% glucose. In addition, 200 μ L each of the diluted suspension were added to eight wells of 96-well sterile polystyrene tissue culture (flat bottom) microtiter plates (Cellstar, Becton Dickinson Labware, Franklin Lakes, NJ), four wells with 200 μ L of medium served as blanks. Ten replica plates were prepared. To minimize evaporative loss, the outermost rows of each plate were filled with 150 μ L of sterile water. After sealing with parafilm, plates were incubated for 48 h at 30°C.

Microtiter Plate Biofilm Assay by Crystal Violet Staining

After incubation, bacterial growth was determined by measuring the OD at 600 nm using a microplate reader (MR600, Dynatech, Lebanon, PA). For assessing the biofilm strength, the method of Djordjevic *et al.* (2002) was used with slight modifications. Briefly, the medium was decanted and unattached cells were removed by rinsing three times with 250 μ L of PBS. Plates were then dried in an inverted position for 30 min at 42°C. Biofilms were stained by adding 150 μ L of aqueous 1% crystal violet solution to each well and by further incubation for 30 min at room temperature. Unbound dye was removed by rinsing three times with 250 μ L of sterile water. The crystal violet was

solubilized by adding 220 μL of 95% ethanol and incubating at 4C for 45 min. A volume of 100 μL from each well were then transferred to a fresh microtiter plate and the OD was read at 595 nm using a microplate reader (MR600, Dynatech Laboratories Inc.).

Quantification of Exopolysaccharide by Ruthenium Red Staining

Microtiter plates with biofilms grown for 48 h at 30C and blank (uninoculated medium) were prepared as described above. After washing, biofilms were stained for exopolysaccharides (EPS) essentially as described by Borucki *et al.* (2003). Briefly, 200 μL of ruthenium red (0.1% in water) was pipetted into each well and left for 45 min at room temperature. The liquid from each well was then transferred to a fresh polystyrene microplate and the OD at 450 nm was measured using a microplate reader (MR600, Dynatech Laboratories Inc.). The amount of dye bound by the biofilm in each well was determined by subtracting the OD measured for the individual well from the average calculated for the blank wells.

Determination of Culture Composition in Mature Dual Species Biofilms

Dual species biofilms of *L. monocytogenes* and *S. epidermidis* with initial mixing ratios of 20:80, 40:60 and 50:50 were grown in 12-well sterile polystyrene tissue culture plates (Cellstar), incubated for 48 h at 30C. Biofilms were washed once with PBS. In addition, 2 mL of PBS per well was added and bacteria were harvested by scraping with a bent tip Pasteur pipette. Complete removal of biofilms was visually checked after crystal violet staining. The harvested cells were vortexed vigorously, serial dilutions were plated onto TSBYE + 1% glucose agar (nonselective), incubated at 37C for 24 h and total plate counts, i.e., for *S. epidermidis* plus *L. monocytogenes*, were determined. After replica plating onto *Listeria* selective medium (PALCAM) and incubation at 37C for 24 h, the *Listeria* plate counts were determined.

Biofilm Growth on Cellulose Dialysis Membranes

This was performed according to Resch *et al.* (2005). Circular sheets of 70–75-mm diameter were cut out from 38-mm wide dialysis tubes (Nadir, pore size 25–30 Å; Roth, Richmond, CA). In order to remove traces of heavy metals and other growth-inhibiting substances, the dialysis membranes were pretreated with 500 mL aqueous 2% sodium bicarbonate/1 mM Na_2 -di-sodium ethylene diamine tetra acetic acid for 3 h at 60C, thoroughly rinsed (8–10 times) in 500 mL of distilled water and stored at 4C. Prior to use, the membranes were sterilized by immersing them in 70% ethanol for 1 h and dried aseptically, then tightly applied, with the help of a glass spatula, on tryptic soy

agar + 1% glucose + 0.6% yeast extract in a standard Petri dish and left at room temperature for 24 h. Furthermore, 200 μ L of 5,000-fold diluted (approximately 5×10^6 cfu/mL) bacterial suspension were evenly plated onto the membranes and the dishes were incubated in an upright position for 48 h at 30C.

Scanning Electron Microscopy

The method of Kalmokoff *et al.* (2001) was used with several modifications. Membranes carrying biofilms were rinsed by gentle repeated immersion in 15–20-mL sterile PBS for 15–30 s, then placed in Petri dishes with 6.25% glutaraldehyde in 50 mM phosphate buffer, pH 7.4 and fixed at 4C overnight. The membranes were rinsed in an excess of 50 mM phosphate buffer and samples were dehydrated with acetone, subjected to critical point drying within a biofilm sputter, coated for 300 s with gold palladium and inspected using a Leo 425 scanning electron microscope (Electron Microscopy Ltd., Cambridge, United Kingdom). Images were captured at low magnification for overview of biofilm structure and high magnification to show details of biofilm 3D-architecture. The images were processed for display using PHOTOSHOP (Adobe Systems Inc., San Jose, CA) software.

Statistical Analysis

All experiments were carried out in triplicate and repeated in three independent sets of experiments. Data were shown as mean $\pm$ standard deviation (SD). SPSS 10.0.5 version for windows (SPSS Software Inc., Chicago, IL) computer program was used for statistical analysis: The significance of differences in biofilm formation was assessed by analysis of variance, followed by post hoc comparison test. Correlations between quantitative properties were evaluated by calculating the Duncan and Dunnett's coefficient. Statistical significance value set at $P < 0.05$.

RESULTS

Growth of mono and dual species cultures in different media

A resuspended preculture was diluted 1:30-fold into four different media (defined HTM, TSB + 1% glucose, TSBYE, TSBYE + 1% glucose) and growth at 30C was monitored for 8 h for monocultures of *L. monocytogenes* (*L. m.*) and *S. epidermidis* (*S. epi.*) and in mixed culture with a ratio of 50:50. With HTM, very slow growth was observed in both mono and mixed cultures, the OD at 600 nm after 8 h reaching just 0.34 ± 0.02 (*L. m.*; mean and SD of

three independent experiments), 0.16 ± 0.01 (*S. epi.*) and 0.23 ± 0.02 (mixed). In TSB + 1% glucose, a monoculture of *L. monocytogenes* grew well (OD 600 1.05 ± 0.02), but this medium failed to support good growth of a monoculture of *S. epidermidis* (OD 600 0.28 ± 0.02) and of the mixed culture (OD 600 0.39 ± 0.02). In TSBYE, the monoculture of *S. epidermidis* grew well (OD 600 1.08 ± 0.05), but both the monoculture of *L. monocytogenes* (OD 600 0.44 ± 0.02) and the mixed culture (OD 600 0.60 ± 0.05) grew considerably less. TSBYE+1% glucose supported the growth of both monocultures (*L. m.* OD 600 0.80 ± 0.01 ; *S. epi.* OD 600 0.84 ± 0.01) and the mixed culture (OD 600 0.83 ± 0.01) to the same extent and therefore was selected for the further experiments. A slightly slower growth of *L. monocytogenes* in TSBYE + 1% glucose when compared to TSB + 1% glucose has previously been observed by us (Zameer *et al.* 2009) and others (Busch and Donnelly 1992).

Quantitative analysis of biofilm formation by mono and dual species cultures

Our previous results (Zameer *et al.* 2009) have shown that 30°C as incubation temperature and 48 h as the incubation time were optimal for the formation of monoculture biofilms by *L. monocytogenes* EGD-e, the same conditions were used in this study. The development of mono and dual species biofilms of *L. monocytogenes* and *S. epidermidis* on polystyrene microtiter plates was monitored by the standard crystal violet assay for adherent cells, expressed as OD at OD 595, which will be referred to as biofilm strength. The monocultures of both the bacteria showed maximum values at 24 h (data not shown), whereas the mixed-type biofilms continued to grow up to 48 h. *S. epidermidis* RP62A formed much stronger biofilms than *L. monocytogenes* EGD-e (Fig. 1A). As described above, the growth rates of these strains were identical in planktonic culture. When assayed after coculture, using different relative amounts of the respective strain but identical total bacterial numbers, complex results were obtained (Fig. 1A). At ratios of *L. monocytogenes* to *S. epidermidis* of 50:50 or 60:40, the biofilm strength was comparable to the pure culture of *S. epidermidis* and significantly more than for pure *L. monocytogenes*. The 80:20 ratio gave the same low value as *L. monocytogenes* alone. The maximum was reached at a ratio of 40:60. When the corresponding amounts of EPS were measured by ruthenium red binding (Fig. 1B), a different pattern was observed. Here, the value for *L. monocytogenes* alone was slightly higher than pure *S. epidermidis* and was rather similar for all ratios of mixture, except for the 40:60 ratio. As observed for biofilm strength, this ratio yielded maximum values for EPS, too.

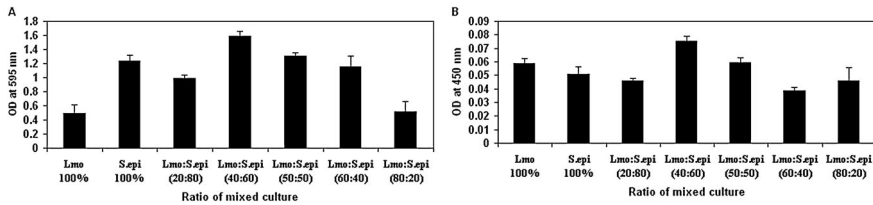


FIG. 1. BIOFILM DEVELOPMENT OF MONO AND DUAL SPECIES CULTURES OF *LISTERIA MONOCYTOGENES* AND *STAPHYLOCOCCUS EPIDERMIDIS* IN MICROTITER PLATES

TSBYE + 1% glucose, pure inocula mixed at different ratios (A): crystal violet staining (OD 595) to evaluate biofilm strength and (B): ruthenium red assay (OD 490) for exopolysaccharide production, after 48 h at 30°C. *L. monocytogenes* 100%, *S. epidermidis* 100%, mixed culture of biofilms in the ratio 20:80, 40:60, 50:50, 60:40 and 80:20 of *L. monocytogenes* and *S. epidermidis*, respectively. The data represent the means and standard deviations of three independent experiments.

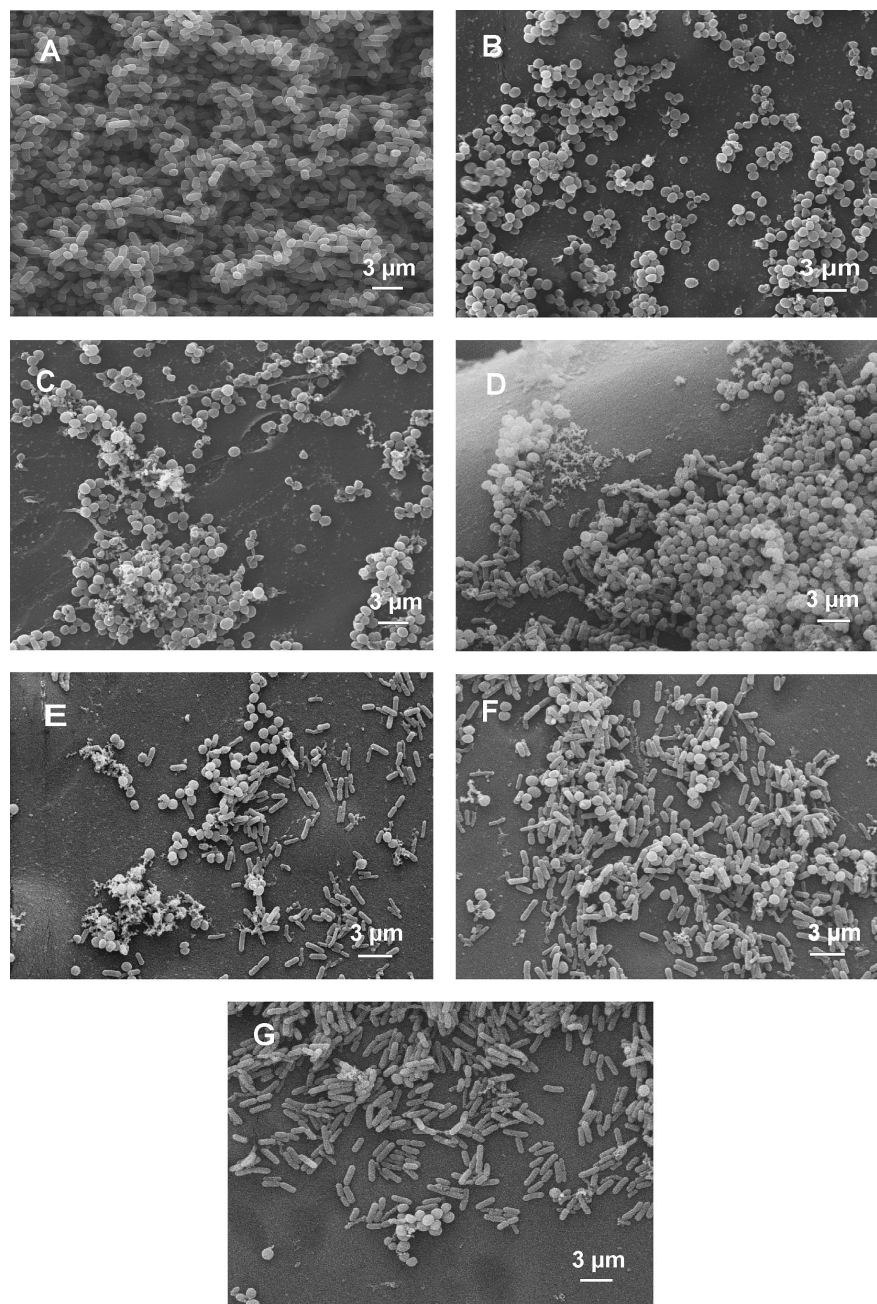
Culture composition in established dual species biofilms

The proportion of *L. monocytogenes* (*L. m.*) and *S. epidermidis* (*S. epi.*), respectively, in established dual species biofilms was measured by determining total (on TSBYE + 1% glucose agar) and *L. monocytogenes* (on selective PALCAM agar) plate counts in biofilms after 48 h of coculture at 30°C in polystyrene plates. When the initial mixing ratio of *L. m.* to *S. epi.* was 20:80, the measured proportion after 48 h was $18.8 \pm 1.7\%$ to $81.2 \pm 8.2\%$ (means and SD from three independent experiments). For an initial ratio of 40:60 (*L. m.* *S. epi.*), the values were $39.3 \pm 2.1\%$ to $60.7 \pm 3.8\%$, and for the 50:50 ratio, $53.5 \pm 6.9\%$ and $46.5 \pm 6.1\%$. Clearly, there was no significant change in the relative amounts of the two bacteria over the incubation time of 48 h.

Scanning electron microscopic analysis of mono and dual species biofilms

In a previous study, we could show that *L. monocytogenes* EGD-e forms high-density biofilms on cellulose dialysis membranes and that biofilms grown on this kind of support are well suited for microscopic examination (Zameer

FIG. 2. SCANNING ELECTRON MICROGRAPHS OF MONO AND DUAL SPECIES CULTURES OF *LISTERIA MONOCYTOGENES* AND *STAPHYLOCOCCUS EPIDERMIDIS* BIOFILMS FORMED ON CELLULOSE DIALYSIS MEMBRANES GROWN AT 30°C ON TSBYE + 1% GLUCOSE OVER A PERIOD OF 48 H AT DIFFERENT RATIOS OF THE MIXED CULTURE INOCULUMS (A) *L. MONOCYTOGENES* 100%, (B) *S. EPIDERMIDIS* 100%, DUAL SPECIES BIOFILMS IN THE RATIO (C) 20:80 (D) 40:60 (E) 50:50 (F) 60:40 AND (G) 80:20 OF *L. MONOCYTOGENES* AND *S. EPIDERMIDIS*, RESPECTIVELY



et al. 2009). Scanning electron microscopy (SEM) of *L. monocytogenes* and *S. epidermidis* biofilms, grown for 48 h at 30°C on cellulose membranes, gave the following results. *L. monocytogenes* in monoculture yielded a dense biofilm, with channels between the aggregated cells and with connecting fibrils (Fig. 2A). *S. epidermidis* formed patchy aggregates in which the individual cells were held together by an amorphous or, in some cases, fibrillar substance (Fig. 2B). In mixed biofilms, the two bacterial species grew closely associated with each other and not in distinct microcolonies (Fig. 2C–G). The relative proportions of *Listeria* and *Staphylococcus* in the biofilms corresponded approximately to the ratio in which they were mixed in the initial inocula. We observed that during the preparation for electron microscopy, a certain amount of the attached bacteria was detached by the washing steps. The biofilm obtained with the ratio of 40:60 proved to be the most stable one; this corresponds to the results from the microtiter plate assay in which this ratio also yielded the most adherent biofilm.

DISCUSSION

In our experiments, a positive interaction between the two bacterial species in mixed culture biofilms was observed. As shown in Fig. 1A, the biofilm strength in a pure culture of *L. monocytogenes* was significantly less than for pure *S. epidermidis*, despite similar growth in the medium used. At ratios where one of the species was dominant (20:80, 80:20), the total biofilm strength was approximately equivalent to the contribution of the respective species, estimated from the values for pure cultures. However, in particular, at the ratios of 40:60 and 50:50, respectively, biofilm strength was not just reflecting the percentage of the individual species but was significantly higher. The determination of the proportion of *L. monocytogenes* to *S. epidermidis* in dual species biofilms after 48 h of incubation showed that it remained the same as in the inoculum. If one assumes that total biofilm strength, measured as OD₅₉₅, was directly proportional to the amount of the respective species in the mixture, theoretical biofilm strength could be calculated as follows. At a ratio of 50:50, the contribution of *L. monocytogenes* would amount to $OD\ 595\ 0.5/2 = 0.25$, that of *S. epidermidis*, to $OD\ 595\ 1.2/2 = 0.6$, and the combined OD₅₉₅ value would be 0.85. But the actually measured value was 1.3 (Fig. 1A). Similar results were obtained for the ratios of 40:60 and 60:40. This strongly suggests a synergistic interaction between the two species when they were present in almost equal amounts and might be related to the composition of the polymeric matrix surrounding the closely associated bacteria. For *L. monocytogenes*, it is evident that the matrix produced by these bacteria contains polysaccharides since it can be stained with ruthenium red, as shown in

Fig. 1B and also by Djordjevic *et al.* (2002). Others could demonstrate that the matrix can be labeled with concavalin A which specifically binds to polysaccharides (Hefford *et al.* 2005). Recently, it has been shown that *L. monocytogenes* EGD-e biofilms also contain extracellular DNA, however, its amount is decreasing at later time points (48 h) of biofilm development (Harmsen *et al.* 2010) as studied here. In case of *S. epidermidis*, it has been shown that the extracellular matrix of biofilms contains, in addition to polysaccharide intercellular adhesin (Mack *et al.* 1994), variable amounts of the accumulation-associated protein Aap (Hussain *et al.* 1997) which will contribute significantly to biofilm strength but will not stain with ruthenium red. This explains our observation that the biofilm strength was less for pure *L. monocytogenes* than for *S. epidermidis* (Fig. 1A), despite the higher amount of matrix polysaccharide found for the former bacterium (Fig. 1B).

A similar study by Rieu *et al.* (2008) describes the interaction in dual species biofilms between *L. monocytogenes* EGD-e and different *S. aureus* isolates. It was shown there that neutral, positive and negative interactions occurred, depending on the particular strain of *S. aureus*. Carpentier and Chassaing (2004) investigated the influence of 29 gram-positive and gram-negative isolates from food processing plants on *L. monocytogenes* biofilm population, wherein 53% of the strains tested had a negative effect, 34% had no effect and 13% had a positive effect, resulting in the increased adhesion of the listerial population. Among the positive isolates was a strain of *Staphylococcus capitis*, a relative of *S. epidermidis*. A positive interaction in dual species bacterial biofilms was also observed when *L. monocytogenes* was cocultured with *Flavobacterium* species (Bremer *et al.* 2001).

The morphological organization reported in the literature as typical for *L. monocytogenes* biofilms (Kalmokoff *et al.* 2001; Borucki *et al.* 2003; Hefford *et al.* 2005) was also observed in our experiments. In our study, SEM of mixed culture biofilm demonstrated that both *L. monocytogenes* and *S. epidermidis* readily integrated in close association in a heterogeneous biofilm structure.

In conclusion, our data confirm that the biotic environment influences the efficacy with which *L. monocytogenes* colonizes surfaces. The presence of microorganisms such as *S. epidermidis* can facilitate the colonization and persistency of *L. monocytogenes* in the environment. Although our experiments were performed in batch culture, it can be assumed that the described interaction between a harmless good biofilm former and a pathogen is also relevant in flow systems, as has previously been suggested by Sasahara and Zottola (1993). The mechanism involved in these interactions remains unknown, but our data suggest that they are complex and it will be interesting to characterize the molecules involved in this interaction. A better understanding of these processes will have a significant impact on the research concerning the spread of infectious microorganisms in general and on food safety.

ACKNOWLEDGMENTS

This work was supported by grants to S.G. from the Department of Science and Technology, Government of India (SR/FT/L-89/2005) and from the Indian Council for Medical Research (5/8/9/56/2006-ECD-I) and to J.K. from the International Bureau/Bundesministerium für Bildung und Forschung, Germany (IND 06/010).

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